

Quiz 4

NAME: _____

Q.1. Solve the inequality $3|3x - 1| - 2 \geq 4$ and write your answer in interval notation.

Q.2. Solve the inequality $\frac{8}{x+3} < 2$ and write your answer in interval notation (**Hint:** start by getting 0 on side of the inequality).

Q.3. Write the **general expression/ formula** for the midpoint of the two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$. Then use this formula to find the midpoint of the points $A(-1, 7)$ and $B(5, 3)$.

Q.4. Determine with working whether or not the point $C(4, 8)$ lies on the perpendicular bisector of the line segment from $A(-1, 7)$ to $B(5, 3)$.